YUEYANG LFG POWER GENERATION PROJECT

Document Prepared by Climate Bridge (Shanghai) Ltd.

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Project Title	Yueyang LFG Power Generation Project
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Yueyang LFG Power Generation Project (hereafter referred to as “the Project”) is a landfill gas (LFG) recovery and utilization project developed by Yueyang Weiteng New Energy Co., Ltd. (hereinafter referred to as “the project proponent”). The Project is located at the west of Yueyang County MSW Landfill, Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People’s Republic of China.

Yueyang County MSW Landfill started operation in August 2013, The MSW deposition rate is estimated to be 350 tons/day during the operational lifespan of 10 years.

Prior to the implementation of the project activity, the LFG generated from Yueyang County MSW Landfill was totally released into the atmosphere, the equivalent amount of electricity generated by the project activity was supplied by the fossil-fuel-dominated Central China Power Grid (CCPG). The baseline scenario is the same as the conditions existing prior to the implementation of the project activity.

The purpose of the Project is to collect LFG from Yueyang County MSW Landfill and utilize it for power generation. It includes LFG collection system, LFG pre-treatment system, flaring and electricity generation system. LFG collected will be used for electricity generation with internal combustion generators. The flare will only be involved when the electricity generation system is not in operation, or the LFG is exceeded. Hence, the project developer determined that VCUs from flare will not be claimed, even if any methane is destroyed by flare during the crediting period. And VCUs will only be claimed from the methane destroyed by electricity generation system and the power displacement. The project emissions from flaring system are excluded for simplification.

The Project started construction on 10-Oct-2020. Two power generators of 1.067 MW each, which add up to 2.134 MW, have been installed, and they were put into operation on 26-Nov-2021.

The Project achieves emission reductions in two aspects: 1) it avoids methane emissions by capturing and destroying LFG which would have been directly vented into the atmosphere in the baseline scenario; 2) it reduces CO_2 emissions by displacing part of the electricity that would otherwise have been supplied by CCPG.

It is estimated that during the 10-year project crediting period (from 26-Nov-2021 to 25-Nov-2031), the total GHG emission reductions will be 285,176 tCO_2e, with annual emission reductions of 28,517 tCO_2e.

Audit Type	Period	Program	VVB Name	Number of years
Validation/ Verification	15-Aug-2023	VCS	Shenzhen CTI International Certification Co., Ltd.	NA

1.2 Sectoral Scope and Project Type

As per the methodology AMS-III.G (version 10.0), if the recovered LFG is used for any purpose other than flaring, then application of sectoral scope 13 and sectoral scope 1 is mandatory.

As the Project uses recovered LFG for power generation, the sectoral scopes applicable are:

- Sectoral scope 1: energy industries (renewable-/non-renewable sources), and
- Sectoral scope 13: waste handling and disposal.

The Project is not a grouped project.

1.3 Project Eligibility

The scope of the VCS Program includes:

1) The seven Kyoto Protocol greenhouse gases: The emissions reduction of the project comes from two sources: 1) Methane (CH_4) emissions from the previously vented landfill gas that will be captured and destroyed in the project scenario; 2) CO_2 emissions from the production of the equivalent amount of electricity replaced by the Project that would otherwise have been provided by the fossil fuel fired power plants of Central China Power Grid (CCPG). Thus, the project applicable to this scope.

2) Ozone-depleting substances: NA

3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: NA

4) Project activities supported by a methodology approved under a VCS approved GHG program unless explicitly excluded under the terms of Verra approval: The methodology AMS-III.G (version 10.0) and AMS-I. D (version 18.0) of the project utilized are methodologies approved under CDM Program, which is a VCS approved GHG program.

5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: NA

The project does not belong to projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.

Meanwhile, the Project does not fall into any project type shown in Table 1 (Excluded Project Activities) of *VCS Standard 4.4*.

Therefore, the Project is eligible under the scope of VCS program.

1.4 Project Design

☒ The project includes a single location or installation only

☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project

☐ The project is a grouped project

Eligibility Criteria

Not applicable as the project has been designed to be a single installation of an activity, not a grouped project.

1.5 Project Proponent

Organization name	Yueyang Weiteng New Energy Co., Ltd.
Contact person	CHENG Xuan
Title	Manager
Address	Room 3102, Building 3, Dongfang Road, Rongjiawan Town, Yueyang Town, Yueyang County, Yueyang City, Hunan province, China
Telephone	+86 -2123019950
Email	3542346576@qq.com

1.6 Other Entities Involved in the Project

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultant
Contact person	GAO Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China
Telephone	+86 021-62462036

Email	projects@climatebridge.com

1.7 Ownership

The ownership of Yueyang Weiteng New Energy Co., Ltd. over the Project arises under the project approval issued by a competent authority in compliance with laws; the ownership also arises by virtue of property rights in the plant and equipment that generate GHG emission reductions, as shown in Table 1.1.

Table 1.1 Evidence establishing project ownership accorded to the project proponent.

Evidence types of project ownership	Condition of the Project and the project proponent
1) Project ownership arising or granted under statute, regulation or decree by a competent authority.	The project proponent has obtained the approval from the Development and Reform Bureau of Yueyang County, for the construction and operation of the Project. The Development and Reform Bureau is a competent authority, and the approval demonstrates that the project proponent has been granted the project ownership in compliance with relevant laws and regulations in China.
2) Project ownership arising under law.	
3) Project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership).	The project proponent has signed equipment purchase contracts with suppliers, and signed the construction contract with contractors; the project proponent has also signed a cooperation agreement with the landfill for the utilization of the landfill gas. Therefore, the project proponent has property rights in the power plant and its equipment that generate GHG emission reductions.

1.8 Project Start Date

The project start date is 26-Nov-2021 when the power generator started operation.

1.9 Project Crediting Period

The Project adopts a fixed ten-year crediting period from 26-Nov-2021 to 25-Nov-2031 (both days included).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO_2e/year

☒ 20,000 – 100,000 tCO_2e/year

☐ 100,001 – 1,000,000 tCO_2e/year

☐ >1,000,000 tCO_2e/year

Project Scale	
Project	✔
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
26-Nov-2021 to 31-Dec-2021	5,059
01-Jan-2022 to 31-Dec-2022	48,817
01-Jan-2023 to 31-Dec-2023	42,097
01-Jan-2024 to 31-Dec-2024	36,421
01-Jan-2025 to 31-Dec-2025	31,620
01-Jan-2026 to 31-Dec-2026	27,550
01-Jan-2027 to 31-Dec-2027	24,094
01-Jan-2028 to 31-Dec-2028	21,154
01-Jan-2029 to 31-Dec-2029	18,646
01-Jan-2030 to 31-Dec-2030	16,501
01-Jan-2031 to 25-Nov -2031	13,217
Total estimated ERs	285,176
Total number of crediting years	10
Average annual ERs	28,517

1.11 Description of the Project Activity

The project activity installs a comprehensive system for LFG recovery and utilization on the existing landfill site. The system involves LFG collection, LFG pre-treatment, flare and electricity generation. The key systems involved in the proposed project are as follows:

- **Gas collection system**

The landfill gas collecting system is a gas pipeline network, consisting of gas collecting wells and one main gas pipe which is connected with each gas collecting well, and gas collection blower. LFG will be extracted by gas collection blower and transported by pipeline from gas collection wells to gas pre-treatment system.

- **Gas pre-treatment system**

Before entering the gas engines, the LFG is pre-treated so that impurities and moisture are removed to avoid corrosion of the electricity generation system. In addition, the pre-treatment system maintains the LFG in a continuously stable condition before the gas generator inlets. The pre-treatment includes filtration, dehumidification, desulfurization, cooling and pressurization.

- **Flare**

The flare will only be involved when the electricity generation system is not in operation, or the LFG is exceeded. Hence, the project developer determined that VCUs from flare will not be claimed, even if any methane is destroyed by flare during the crediting period. And VCUs will only be claimed from the methane destroyed by electricity generation system and the power displacement. The project emissions from flaring system are excluded for simplification.

- **Electricity generation system**

2 gas generators with a rated capacity of 1.067 MW each have been installed and they are fed with the LFG to generate electricity, which is then exported to the grid.

The service provided by the Project is electricity. In the baseline scenario, the same amount of electricity would have been supplied by power plants connected to the grid.

The detailed information of the Gas Generator of the project is shown in the table below.

Table 1.2 Technical parameters of the gas generator[1]

Manufacturer	JENBACHER
Type	JGS320GS-B.L.
Quantity	2

[1] Sourced from gas generator nameplate.

Rated capacity	1.067 MW
Rated power factor	1
Rated frequency	50 HZ
Rated voltage	400 V
Rated current	1,540 A

1.12 Project Location

The Project is located in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People's Republic of China. The coordinates of the project site are E 113°11'24", N 29°11'13".

The project location is shown in Figure 1.1.

Figure 1.1 Location of the project

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity is:

1) LFG captured and utilized by the project was released directly into the atmosphere. And,

2) the quantity of electricity supplied by the project would have been produced by power plants connected to the power grid (CCPG).

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project complies with all Chinese relevant laws and regulations, as shown in Table 1.2.

Table 1.2 Compliance with relevant laws and regulations

Laws and regulations	The Project
Regulations on the Approval and Recordation of Enterprise Investment Projects[2], Administrative Measures on the Approval and Recordation of Enterprise Investment Projects[3], which sets out the procedures of project approval and recordation.	The Project has obtained the approval and has been recorded by the Development and Reform Bureau of Yueyang County, in compliance with the regulation and the administrative measure.
Law of People's Republic of China on Environmental Impact Assessment[4], which sets out the requirements on the completion and approval of the environmental impact assessment (EIA) report/form of construction projects.	The (EIA) form of the Project has been completed and then approved by the Yueyang County Branch of Yueyang Ecology and Environment Bureau, which is in compliance with the provisions in the law.
Construction Law of the People's Republic of China[5], which sets out the requirements on application and approval of the construction permit prior to the project construction.	The Project obtained the construction permit prior to the construction in compliance with the provisions in the law.
Catalogue for the Guidance of Industrial Structure Adjustment[6], which lists projects in	The Project is a landfill gas project, and belongs to the encouragement category of the Catalogue.

[2] http://www.gov.cn/zhengce/content/2016-12/14/content_5147959.htm

[3] https://www.ndrc.gov.cn/xxgk/zcfb/fzggwl/201703/W020190905495074985482.pdf

[4] https://www.mee.gov.cn/ywgz/fgbz/fl/201901/t20190111_689247.shtml

[5] http://www.npc.gov.cn/npc/c30834/201905/0b21ae7bd82343dead2c5cdb2b65ea4f.shtml

[6] http://www.gov.cn/xinwen/2019-11/06/5449193/files/26c9d25f713f4ed5b8dc51ae40ef37af.pdf

three categories: encouragement category, restriction category and elimination category.	

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The Project has neither been registered nor seeking registration under any other GHG programs.

The Project is seeking registration only under VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The Project has not been rejected by any GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

Supply Chain (Scope 3) Emissions

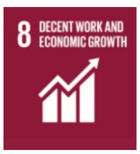
The project is an LFG power generation project, which reduces GHG emission reductions by avoiding CO_2 emissions from those fossil fuel-based power plants connected to the grid and by avoiding GHG emissions from releasing LFG into atmosphere at the landfill site. The project proponent is producer of the electricity generated from LFG, the owner of the recovered LFG is Yueyang County MSW Landfill which provides waste management service. The project proponent promised that the emission reduction generated by the project will only seek VCUs and informed orally Yueyang County MSW Landfill that the project will not be double counting.

As per *VCS Standard 4.4*., public statements are required for all emission reductions associated with renewable electricity for which VCUs may be requested from 01-Jul-2023 onwards. Therefore, the project is not applicable to this section.

1.17 Sustainable Development Contributions

The Project contributes to achieving sustainable development goals in the following aspects:

The project is to capture and utilize landfill gas for power generation to avoid landfill gas emissions and CO_2 emissions from the production of the equivalent amount of electricity replaced by the project that would otherwise have been purchased from the CCPG. The project will achieve a GHG emission reduction of 28,517 tCO_2e/yr during the crediting period. Thus, the project will achieve SDG 13 Climate Action[7]. This contributes to achieve China's stated sustainable development priorities "China's carbon dioxide emissions would strive to peak by 2030 and strive to achieve carbon neutrality by 2060".



The construction and operation of the project will provide new local jobs and increase tax revenue, which will have a positive effect on the local economy. During the crediting period, direct and indirect employment opportunities will be generated. Thus, the project will achieve SDG 8 Decent Work and Economic Growth[8]. This contributes to one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value".

Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in Central China Power Grid and further help to achieve the national action to promote renewable energy development. Thus, the project will achieve SDG 7 Gender Equality[9]. This contributes to achieve China's stated sustainable development priorities "By 2030, the proportion of non-fossil energy consumption will reach about 25%".

[7] https://sdgs.un.org/goals/goal13

[8] https://sdgs.un.org/goals/goal8

[9] https://sdgs.un.org/goals/goal7

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project is not a AFOLU project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable as there is no further information relevant to the project.

2 SAFEGUARDS

2.1 No Net Harm

The Environmental Impact Assessment (EIA) form for the Project has been completed by a third-party company (Please refer to Section 2.3 for details), and approved by the Yueyang County Branch of Yueyang Ecology and Environment Bureau, with approval Yue Xian Huan Ping Pi No. (2020) No. 57. Every aspect of environmental impacts has been considered and discussed in the EIA form; with necessary mitigation measures taken, there are no negative environmental impacts due to the Project.

Socially and economically, the Project brings many benefits to local people, as mentioned in Section 1.17. The Project does not result in any negative socio-economic impacts.

In conclusion, the construction and operation of the Project does not result in net environmental or socio-economic harm.

2.2 Local Stakeholder Consultation

Local stakeholder consultation before the Project construction

The local stakeholders identified include personnel of the landfill and the power plant, local residents around the project area and government officials. Prior to the Project construction, the project proponent conducted local stakeholder consultation by distributing survey questionnaires in November 2020, to get informed of the stakeholders' opinion about the Project in various aspects. The surveyed stakeholders varied in age, gender, occupation and educational background, as shown in Table 2.1.

Table 2.1 Basic information of the stakeholders surveyed

Item	Distribution	Number	Percentage
Gender	Male	25	50.0%
	Female	25	50.0%
Age	<25	8	16.0%
	25-45	20	40.0%
	45-60	12	24.0%
	>60	10	20.0%
Education	Junior high school or below	24	48.0%
	Senior high school	15	30.0%
	College or above	11	22.0%
Occupation	Worker	20	40.0%
	Farmer/peasant	3	6.0%
	Management personnel	8	16.0%
	Civil servant	10	20.0%
	Other	9	18.0%

A total number of 50 questionnaires were distributed, and all questionnaires were recollected. The survey results are summarized in Table 2.2.

The survey shows that the Project is well known by local residents. Most of them are supportive of the project construction and believe that the Project will bring more benefit than loss. 84.0% of the respondents have confirmed their willingness to participate in the construction and/or operation of the Project. 86% of the respondents believed that the Project will not have any environmental impact, but a few respondents showed their concern about the potential negative environmental impacts caused by the Project. Regarding the landfill condition, 96% of the respondents considered that the Project will have positive impacts. All the respondents believe that the Project will create employment opportunities; almost all of them believe that the Project will contribute to local economic development and increase local power supply.

Table 2.2 Summary of the survey results

No.	Questions	Response	Number	Percentage
1	Are you informed of the project activity?	Very much	44	88.0%
		A little	6	12.0%
		Not at all	0	0%
2		Supportive	46	92.0%

	In general, what is your attitude towards the project construction?	Against	0	0%
		Indifferent	4	8.0%
3	What do you think will be the major environmental impacts of the project? (multiple choices possible)	None	43	86.0%
		Air pollution	3	6.0%
		Water pollution	2	4.0%
		Noise pollution	2	4.0%
		No idea	0	0%
4	Do you believe that the project operation will bring more benefit than loss?	Yes	50	100%
		No	0	0%
		No idea	0	0%
5	Are you willing to participate in the project construction and/or operation?	Yes	42	84.0%
		No	6	12.0%
		Indifferent	2	4.0%
6	What influence do you think the project will bring to you and your family?	Positive	46	92.0%
		None	4	8.0%
		Negative	0	0%
7	What kind of effect do you think the project operation will have on the landfill condition?	Positive effect	48	96.0%
		None	2	4.0%
		Negative effect	0	0%
8	What kind of impact do you think the project will bring to the local employment?	Increase	50	100%
		Decrease	0	0%
		No impact	0	0%
9	What kind of impact do you think the project will bring to the local economy?	Positive	48	96.0%
		Negative	2	4.0%
		No impact	0	0%
10	What kind of impact do you think the project will bring to the local power supply?	Increase	46	92.0%
		Decrease	0	0%
		No impact	4	8.0%

In conclusion, the implementation of the Project is regarded as beneficial by majority of the local stakeholders.

Local stakeholder consultation during the project operation

During the project implementation, local stakeholders' opinions are collected through two channels available: regular questionnaire surveys conducted by the project proponent which take place every year, communications between local residents and authorities. The project proponent informs the local authorities of key implementation events or changes of the Project, then the local authorities inform the residents living around the project sites, and the comments and suggestions from residents are collected by the local authorities; the local government agencies also conduct spot checks on the project implementation on a regular basis, and give suggestions on potential issues.

2.3 Environmental Impact

The project proponent entrusted a third party, Hunan Tenghui Environmental Protection Consulting Service Co., Ltd., to conduct the Environmental Impact Assessment (EIA) on the Project. The EIA form, completed by the third party, has been approved by the Yueyang County Branch of Yueyang Ecology and Environment Bureau. The environmental impacts and associated mitigation measures are summarized as follows.

1. Construction Period

1.1 Air pollution

Major impact on air quality during the construction period include dust caused by machinery use and by transport of construction materials and waste. Appropriate measures should be taken to minimize the impacts, such as enclosure, watering, transportation vehicle cleaning, etc.

1.2 Wastewater

During construction, the waste water mainly comes from construction water. The construction wastewater is reused after treatment and will not be discharged.

1.3 Noise

Noise during the construction period is mainly generated by the construction machineries and vehicles. To minimize noise pollution, optimize the construction layout, strictly control the construction period, and avoid the impact of noise on the environment during the construction period.

1.4 Solid waste

The solid waste during the construction period is mainly construction waste. Construction waste shall be cleaned and transported in a timely manner. If it is stored in the site, it shall be covered with airtight dustproof net.

2. Operation Period

2.1 Air pollution

Standardize the construction of landfill gas pretreatment devices, and the landfill gas will enter the generator unit after desulfurization, desilication, purification and other pretreatment. The tail gas of the generator is discharged through the 8m exhaust funnel after being purified by denitration, and the discharged exhaust gas complies with the requirements of the special emission limit standard in Table 3 of *Boiler Air Pollutant Emission Standard* (GB13271-2014).

2.2 Wastewater

Complete the construction of rainwater pipe network in the plant area in strict accordance with the principle of "rainwater and sewage diversion and sewage diversion". The rainwater enters the rainwater pipeline of the landfill through the rainwater pipeline of the plant and finally enters the Xinqiang River; Domestic sewage and condensate will be sent to Xinqiang River after being treated by the leachate treatment station of Yueyang Domestic Waste Landfill. According to the principle of zoning prevention and control, do a good job in the sealing and corrosion prevention of sewage pipe network and the seepage prevention and leakage prevention of the temporary storage site of hazardous waste to prevent the leakage liquid from polluting the regional groundwater.

2.3 Noise

The reasonable layout of the project, the selection of low-noise equipment and the adoption of measures such as shock absorption, sound insulation and noise elimination ensure that the noise at the factory boundary during the operation period of the project meets Category 2 requirements in the national standard *Emission Standard for Industry Enterprise Noise at Boundary (GB 12348-2008)*.

2.4 Solid waste

According to the *Standards for Pollution Control on the Storage and Disposal Site for General Industrial Solid Wastes (GB18599-2001)*, the 2013 revision list of *(GB18599-2001)*, *Standard for Pollution Control on Hazardous Waste Storage (GB18596-2001)* and the relevant requirements of the revision list, the temporary storage sites for general industrial solid waste and hazardous waste are set up in a standardized manner. Establish a management account for the whole process of temporary storage, transfer and disposal of solid waste, and properly collect waste engine oil and catalyst and send them to qualified units for disposal; The replaced waste adsorbent and waste filter element shall be recycled by the manufacturer; General industrial solid waste such as waste residue from landfill gas purification and treatment will be collected and treated by the environmental sanitation department.

In conclusion, the environmental impacts during the construction are temporary and not significant, and the environmental impacts during the operation are minor. The project proponent will take appropriate measures to minimize these negative environmental impacts.

2.4 Public Comments

The Project will be listed on the Verra website for public comment.

2.5 AFOLU-Specific Safeguards

Not applicable as the project is non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Project applies the small-scale methodology AMS-III.G Landfill methane recovery (version 10.0) and AMS-I.D Grid-connected renewable electricity generation (Version 18.0). Please refer to the following link for details:

https://cdm.unfccc.int/methodologies/DB/0KHNES8D09H134V3TZDQ47C3LQL3H2

https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK

The methodologies also refer to the latest approved versions of the following tools:

TOOL04: “Emissions from solid waste disposal sites” (version 08.0):

https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-04-v8.0.pdf

TOOL07: “Tool to calculate the emission factor for an electricity system” (version 07.0):

https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

TOOL32: “Positive lists of technologies” (version 04.0):

https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-32-v4.0.pdf

3.2 Applicability of Methodology

The Project fulfils each of the applicability conditions of the applied methodology AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0) as shown in Table 3.1.

Table 3.1 Applicability of the Project to AMS-III.G.and AMS-I.D

Applicability conditions of AMS-III.G. (version 10.0)	**Justifications**
This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.	Applicable. The project is to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.

Different options to utilize the recovered landfill gas as detailed in paragraph 4 of "AMS-III.H.: Methane recovery in wastewater treatment" (version 19.0) are eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard.	Applicable. The project is to capture and utilize the landfill gas for electrical energy generation directly, which falls under 4(a) of para.4 of AMS-III.H. And the LFG power generation component of the project shall apply methodology AMS-I.D.
Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity	Applicable. Expected aggregate emission reductions from all type III components under the project activity are less than 60 kt CO_2 equivalent annually. The project's average annual emission reduction is 28,517 tCO_2e, which is less than 60,000 tCO_2e.
The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity.	Applicable. The project does not impact the management of the landfill, also does not reduce the amount of organic waste that would have been recycled in the absence of the project activity.
This methodology is not applicable if the management of the solid waste disposal site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to an SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS, or changing the shape of the SWDS to increase methane production.	N.A. The management of the SWDS in the project activity will not be deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity.
Applicability conditions of AMS-I.D. (version 18.0)	**Justifications**
This category comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal, and renewable	Applicable. The project will utilize the LFG to supply electricity to Central China Power Grid.

biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	
This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	Applicable. The project installs a Greenfield LFG power plant at the existing Yueyang County MSW Landfill where there was no renewable energy power plant operating prior to the implementation of the project activity.
Hydropower plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m^2; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m^2.	N.A. The project is not a hydropower plant.
If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel5, the capacity of the entire unit shall not exceed the limit	N.A. The new unit has only renewable components, and the installed capacity of the gas engines is 2.134 MW, significantly less than 15MW.

of 15 MW.	
Combined heat and power (co-generation) systems are not eligible under this category.	N.A. The project does not involve a co-generation system.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	N.A. The project does not involve the capacity addition at an existing renewable power generation facility.
In the case of retrofit, rehabilitation, or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	N.A. The project does not involve retrofit, rehabilitation, or replacement
In the case of landfill gas, waste gas, wastewater treatment, and agro-industries projects recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with the procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	Applicable. The project recovers LFG for electricity generation. Hence recovered methane emissions are eligible under Type III methodology AMS-III.G (version 10.0). And the baseline for the electrical component is in accordance with the procedure prescribed in AMS-I.D (version 18.0).
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply	N.A. The project does not involve biomass sourced from dedicated plantations.

The Project also meets each of the applicability conditions of the applied methodological tools as shown in Table 3.2.

Table 3.2 Applicability of the Project to the applied methodological tools

TOOL04: "Emissions from solid waste disposal sites" (version 08.0)	
Applicability conditions	**Justifications**

The tool can be used to determine emissions for the following types of applications: (a) Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. “ACM0001: Flaring or use of landfill gas”). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g., measuring the amount of methane captured from the SWDS); (b) Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	Application A applies: the project activity mitigates methane emissions from Yueyang County MSW Landfill by capturing and utilizing the methane for power generation. This tool is only applied for ex ante estimation of the emissions in the PD.
TOOL07: “Tool to calculate the emission factor for an electricity system” (version 07.0)	
Applicability conditions	**Justifications**
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	Applicable. The project activity utilizes recovered LGF to export electricity to the grid, and thus substitutes grid electricity. Therefore, this tool is applied to estimate the OM, BM and CM for baseline emissions.

Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e., option II a and option II b. If option II a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	As off-grid power generation is an insignificant part of the national energy mix in China, the emission factor for the project electricity system is calculated only for the grid power plants, which satisfies this applicability condition.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Applicable. The project electricity system is totally located in China, a non-Annex I country.
Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The calculation of the emission factor considers the emission factor of biofuels as zero.
TOOL32: "Positive lists of technologies" (Version 04.0)	
Applicability conditions	**Justifications**
The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM PoA for demonstrating their additionality.	Applicable. The project selects to use the tool to demonstrate the additionality.
This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable. The Project applies this methodological tool in conjunction with the small-scale methodology AMS-III.G. (version 10.0).
The positive lists as contained in section 5 of this tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and	Applicable. The positive lists are valid at the time of

previous versions of methodologies and methodological tools in the "Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools", there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.	the PD writing. Prior to the implementation of the project activities, the landfill gas (LFG) was only vented, but not utilized for energy generation, and that under the project activity, the LFG is used to generate electricity in one power plant with a total nameplate capacity of 2.134MW.

3.3 Project Boundary

According to the methodology AMS-III.G (Version 10.0), the project boundary is the physical, geographical site of the Landfill where the LFG is captured and destroyed/used, and as per AMS-I.D (Version 18.0), the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to.

According to the actual implementation of the project activity, the project boundary includes the Yueyang County MSW Landfill, the whole LFG-related system (e.g. LFG collection, LFG pre-treatment system, LFG power generation system, etc.), and all grid-connected power plants in CCPG.

According to authoritative documents provided by China DNA, CCPG covers Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province, and Chongqing City. The project boundary is shown in Figure 3.1 below.

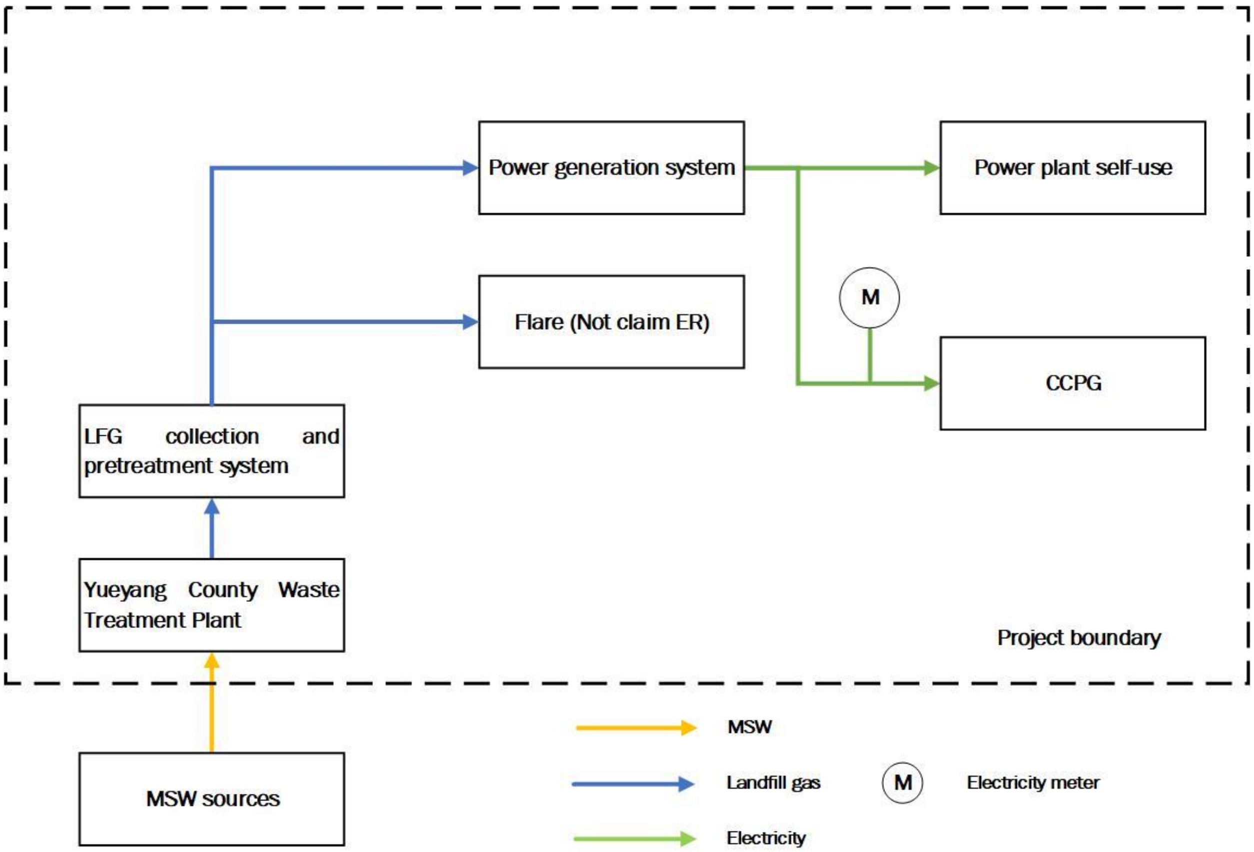


Figure 3.1 Diagram of the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO_2	No	CO_2 emissions from decomposition of organic waste are not accounted since the CO_2 is also released under the project activity.
		CH_4	Yes	The major source of emissions in the baseline.
		N_2O	No	N_2O emissions are small compared to CH_4 emissions from SWDS. This is conservative.
	Emissions from electricity generation	CO_2	Yes	Major emission source, given that power generation is included in the project activity.
		CH_4	No	Excluded for simplification. This is conservative.

	Source	Gas	Included?	Justification/Explanation
		N_2O	No	Excluded for simplification. This is conservative.
	Emissions from heat generation	CO_2	No	The project activity does not involve heat generation.
		CH_4	No	The project activity does not involve heat generation.
		N_2O	No	The project activity does not involve heat generation.
	Emissions from the use of natural gas	CO_2	No	The project activity does not involve the use of natural gas.
		CH_4	No	The project activity does not involve the use of natural gas.
		N_2O	No	The project activity does not involve the use of natural gas.
Project	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO_2	No	The project activity does not involve fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity.
		CH_4	No	The project activity does not involve fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity.
		N_2O	No	The project activity does not involve fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity.
	Emissions from electricity consumption due to the project activity	CO_2	Yes	May be an important emission source.
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small.
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from flaring	CO_2	No	Excluded for simplification, Emissions are considered negligible

Source		Gas	Included?	Justification/Explanation
		CH_4	No	Excluded for simplification[10]
		N_2O	No	Excluded for simplification, Emissions are considered negligible
	Emissions from distribution of LFG using trucks and dedicated pipelines	CO_2	No	The project activity does not involve distribution of LFG using trucks or dedicated pipelines.
		CH_4	No	The project activity does not involve distribution of LFG using trucks or dedicated pipelines.
		N_2O	No	The project activity does not involve distribution of LFG using trucks or dedicated pipelines.

3.4 Baseline Scenario

As per AMS-III.G (Version 10.0), the baseline scenario is the situation where, in the absence of the project activity, biomass and other organic matter are left to decay within the project boundary, and methane is emitted to the atmosphere, possibly with capture of LFG and destruction through flaring to comply with regulations or contractual requirements.

As per AMS-I.D (Version 18.0), the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Therefore, the baseline scenario for the project activity is in the absence of the project activity, LFG from Yueyang County MSW Landfill is emitted directly to the atmosphere, and the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

3.5 Additionality

To demonstrate additionality, the project proponent applies the simplified procedures in Section 5.2 of the methodology AMS-III.G (Version 10.0), according to which, the methodological tool "TOOL32: Positive lists of technologies" (version 04.0) shall be referenced.

The project activity exclusively applies the technology listed in Section 5.1.1 (landfill gas recovery and its gainful use) of TOOL32 (version 04.0) and it fulfils the related conditions as shown in Table 3.2. Therefore, the project activity is deemed automatically additional.

[10] The flare will only be involved when the electricity generation system is not in operation, hence the project developer determined that VERs from flare will not be claimed, even if any methane is destroyed by flare during the crediting period. VERs will only be claimed from the methane destroyed by power generation system and the power displacement.

Table 3.2 Applicability of the Project to TOOL32

Section 5.1.1 of TOOL32 (version 04.0)	The Project
Landfill gas recovery and its gainful use The project activities and PoAs at new or existing landfills (greenfield or brownfield) are deemed automatically additional if it is demonstrated that prior to the implementation of the project activities and PoAs the landfill gas (LFG) was only vented and or flared (in the case of brownfield projects)or would have been only vented and/or flared(in the case of greenfield projects) but not utilized for energy generation, and that under the project activities and PoAs any of the following conditions are met: (a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW; (b) The LFG is used to generate heat for internal or external consumption; (c) The LFG is flared.	The Project involves landfill gas recovery and its gainful use. Prior to the implementation of the project activity, the LFG from Yueyang County MSW Landfill was vented to the atmosphere but not utilized for energy generation. The project activity utilizes LFG to generate electricity and the total nameplate capacity of all generators installed in the project is 2.134 MW, which is below 10 MW.

The additionality of the Project has thus been demonstrated.

3.6 Methodology Deviations

As plant specific fuel consumption and electricity generation data are not publicly available in China, the Guidance caused by DNV's request for deviation of Chinese project activities for the CDM baseline methodology AM0005 has been applied for calculating the build margin (BM) emission factor of this project activity. Please refer to section 4.2 of this report for details.

The deviation has no impact on the conservativeness of the quantification of GHG emission reductions or removals.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The emission reductions are calculated in accordance with AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0). The baseline emissions are determined according to equation (1) and comprise the following sources:

$$BE_y = BE_{CH4,y} + BE_{EC,y} \quad (1)$$

Where:

BE_y	=	Baseline emissions in year *y* (tCO_2e)
$BE_{CH4,y}$	=	Baseline emissions of methane from the SWDS in year *y* (tCO_2e)
$BE_{EC,y}$	=	Baseline emissions associated with electricity generation in year *y* (tCO_2)

As per AMS-III.G (Version 10.0), Baseline emissions of methane from the SWDS ($BE_{CH4,y}$) is calculated ex-ante using equation (2) below:

$$BE_{CH4,y} = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} \times GWP_{CH4} \quad (2)$$

Where:

$BE_{CH4,y}$ = Baseline emissions of methane from the SWDS in year *y* (tCO_2e/yr)

$BE_{CH4,SWDS,y}$ = Methane emission potential of a solid waste disposal site (in tCO_2e) calculated using the methodological tool "Emissions from solid waste disposal sites". This tool may be used:

- With the factor "*f*=0.0" because the amount of LFG that would have been captured and destroyed is already accounted for in this equation.
- With the definition of year x as 'the year since the landfill started receiving wastes, x runs from the first year of landfill operation (*x*=1) to the year for which emissions are calculated (*x*=*y*)'.

The amount of waste type *j* deposited each year *x* ($W_{j,x}$) shall be determined by sampling (as specified in the above-mentioned tool), in the case that waste is generated during the crediting period. Alternatively, for existing SWDS, if the pre-existing amount and composition of the wastes in the landfill are unknown, they can be estimated by using parameters related to the serviced population or industrial activity, or by comparison with other landfills with similar conditions at regional or national level

OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering the waste) (dimensionless). A default value of 0.1 may be used

η_{PJ}	=	Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex-ante estimation only. A default value of 50 percent may be used
$F_{CH4,BL,y}$	=	Methane emissions that would be captured and destroyed to comply with national or local safety requirements or legal regulations in the year *y* (tCH_4). The relevant procedures in "ACM0001: Flaring or use of landfill gas" may be followed, as well as taking into account the compliance with the relevant local laws and regulations if such laws and regulations exist
GWP_{CH4}	=	Global warming potential of CH_4 (tCO_2e/tCH_4)

$BE_{CH4,SWDS,y}$ is determined using the methodological tool "Emissions from solid waste disposal sites" (version 08.0). The following guidance should be taken into account when applying the tool:

(a) f_y in the tool shall be assigned a value of 0 because there's no LFG recovery and utilization facility in the baseline;

(b) In the tool, *x* begins with the year that the SWDS started receiving wastes (e.g. the first year of SWDS operation); and

(c) Sampling to determine the fractions of different waste types is not necessary because the waste composition can be obtained from previous studies.

The $BE_{CH4,SWDS,y}$ is calculated as follows:

$$BE_{CH4,SWDS,y} = \varphi_y \times (1-f_y) \times GWP_{CH4} \times (1-OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^{y}\sum_{j}\left(W_{j,x} \times DOC_j \times e^{-k_j \cdot (y-x)} \times (1-e^{-k_j})\right) \quad (3)$$

Where:

$BE_{CH4,SWDS,y}$	=	Baseline methane emissions occurring in year *y* generated from waste disposal at an SWDS during a time period ending in year *y* (tCO_2e/yr)
x	=	Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (*x* = 1) to year *y* (*x* = *y*)
y	=	Year of the crediting period for which methane emissions are calculated (*y* is a consecutive period of 12 months)
$DOC_{f,y}$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year *y* (weight fraction)
$W_{j,x}$	=	Amount of solid waste type *j* disposed or prevented from disposal in the SWDS in the year *x* (t)
φ_y	=	Model correction factor to account for model uncertainties for year *y*

f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year *y*
GWP_{CH4}	=	Global Warming Potential of methane
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of methane in the SWDS gas (volume fraction)
MCF_y	=	Methane correction factor for year *y*
DOC_j	=	Fraction of degradable organic carbon in the waste type *j* (weight fraction)
k	=	Decay rate for the waste type *j* (1/yr)
j	=	Type of residual waste or types of waste in the MSW

Determination of $\boldsymbol{F_{CH4,BL,y}}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively referred to as required in this section). The four cases in Table 4.1 are distinguished. The appropriate case should be identified, and the corresponding instructions followed.

Table 4.1 Case for determining methane captured and destroyed in the baseline

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

There was no regulation or standard that enforces methane destruction in LFG when Yueyang County MSW Landfill started operation. Furthermore, Yueyang County MSW Landfill did not have existing LFG capture and destruction system prior to the implementation of the project activity. Therefore, this project is in line with case 1: No requirement to destroy methane exists, and no existing LFG capture system.

In this situation:

$$F_{CH4,BL,y} = 0$$

As per AMS-I.D (version 18.0), Baseline emissions associated with electricity generation ($BE_{EC,y}$) is calculated using equation (4) below.

$$BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y} \quad (4)$$

$BE_{EC,y}$	=	Baseline emissions associated with electricity generation in year y (tCO_2)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh)
$EF_{grid,y}$	=	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/MWh)

For greenfield project,

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad (5)$$

Where,

$EG_{PJ,facility,y}$	=	Quantity of net electricity generation supplied by the project plant/unit to the grid in year *y* (MWh)

Determination of the emission factor for electricity generation ($EF_{grid,y}$ = $EF_{grid,CM,y}$)

The combined margin emission factor of the grid ($EF_{grid,CM,y}$) is calculated using the procedures in TOOL07 (version 07.0), and the following six steps are applied:

Step 1: Identify the relevant electricity systems;

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);

Step 3: Select a method to determine the operating margin (OM);

Step 4: Calculate the operating margin emission factor according to the selected method;

Step 5: Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor.

<u>Step 1: Identify the relevant electricity systems.</u>

The delineation of the electricity systems in China is provided by the Development and Reform Commission of China. Among the six regional power grids, Central China Power Grid (CCPG) covers Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province and Chongqing City. Since the Project is located in Hunan Province, CCPG is identified as the relevant electricity system.

<u>Step 2: Choose whether to include off-grid power plants in the project electricity system (optional).</u>

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Considering the structure of China's power system, only grid power plants are included in the calculation.

Step 3: Select a method to determine the operating margin (OM).

The calculation of the operating margin emission factor $EF_{grid,OM,y}$ is based on one of the following methods:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch data analysis OM, or

(d) Average OM.

In China, detailed data from each power plant are sensitive business information and are mostly confidential and thus not publicly available. Therefore, method (b) and method (c) are not suitable for the calculation.

The simple OM method can be used if low-cost/must-run resources[11] constitute less than 50% of total grid generation in average of the five most recent years. According to *China Electric Power Yearbook* released from 2014 to 2018, for CCPG to which the project activity is connected, the low-cost/must-run power generation accounted for 39.13%, 45.07%, 47.01%, 49.05% and 49.08% of total grid generation in 2013, 2014, 2015, 2016 and 2017, respectively; considering the average of the five years from 2013 to 2017, the low-cost/must-run power generation accounted for 45.87% of total grid generation, lower than 50%. Therefore, method (a) is applicable, and the simple OM method is applied for the calculation of the operating margin emission factor $EF_{grid,OM,y}$.

For the simple OM, the emissions factor can be calculated using either of the two following data vintages:

(a) Ex ante option: if the ex ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD to the DOE for validation. For

[11] Low-cost/must-run resources are defined as power plants with low marginal generation costs or power plants that are dispatched independently of the daily or seasonal load of the grid. They typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation. If coal is obviously used as must-run, it should also be included in this list.

off-grid power plants, use a single calendar year within the five most recent calendar years prior to the time of submission of the PD for validation;

(b) Ex post option: if the ex post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required to calculate the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year (y–1) may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year proceeding the previous year (y–2) may be used. The same data vintage (y, y–1 or y–2) should be used throughout all crediting periods.

Based on the most recent data available at the time of this project description submission, the first option (ex-ante) for the calculation of the OM emission factor is selected for the project, and $EF_{grid,OM,y}$ is fixed during the crediting period. The generation data for various power generating sources for the most recent three years (2015-2017) are applied.

Step 4: Calculate the operating margin emission factor according to the selected method.

The simple OM method is applied to calculate the OM emission factor, which is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. It may be calculated by one of the two following options:

Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit; or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:

(i) The necessary data for Option A is not available; and

(ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(iii) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2)

The data of each power plant serving the system are difficult to obtain; in this case, Option A is not preferred. In addition, according to *China Energy Statistical Yearbook*, only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; also, off-grid power plants are not included in the calculation, as discussed in Step 2), which justifies the applicability of Option B for the calculation of the OM emission factor.

Under this option, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i (FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y})}{EG_y} \quad (6)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year *y* (tCO_2/MWh)
$FC_{i,y}$	=	Amount of fuel type *i* consumed in the project electricity system in year *y* (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type *i* in year *y* (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type *i* in year *y* (tCO_2/GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year *y* (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year *y*

If available, values of $NCV_{i,y}$ and $EF_{CO2,i,y}$ provided by the fuel supplier of the power plants in invoices may be used; otherwise, regional or national average default values may be used. For the Project, the values of $NCV_{i,y}$ for each type of fuel are obtained from *China Energy Statistical Yearbook 2018*, and the emission factors $EF_{CO2,i,y}$ for each type of fossil fuel come from default values in *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Fuel consumption data and electricity generation data are obtained from *China Electric Power Yearbook* 2016~2018 and *China Energy Statistical Yearbook* 2016~2018.

The calculation result of the OM emission factor $EF_{grid,OMsimple,y}$ for CCPG is 0.8587 tCO_2/MWh, as is confirmed by *2019 Baseline Emission Factors for Regional Power Grids in China* published by the Ministry of Ecology and Environment of China.

<u>Step 5: Calculate the build margin (BM) emission factor.</u>

As per Section 6.5 of TOOL07 (version 07.0), in terms of vintage of data, project participants can choose between one of the following two options:

Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Option 1 is chosen; the BM emission factor is calculated ex ante for the first crediting period based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5-units}$) and determine their annual electricity generation ($AEG_{SET-5-units}$, in MWh);

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\ percent}$) and determine their annual electricity generation ($AEG_{SET \geq 20\ percent}$, in MWh);

(c) From $SET_{5-units}$ and $SET_{\geq 20\ percent}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. In this case ignore Steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activities, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample-CDM}$) the annual electricity generation ($AEG_{SET-sample-CDM}$, in MWh); if the annual electricity generation of that set comprises at least 20% of the annual electricity generation of the proposed project electricity system (i.e. $AEG_{SET-sample-CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample-CDM}$ to calculate the build margin. Ignore steps (e) and (f).

Otherwise:

(e) Include in the sample group $SET_{sample-CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample-CDM->10yrs}$).

The BM emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units *m* during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m (EG_{m,y} \times EF_{EL,m,y})}{\sum_m EG_{m,y}} \quad (7)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year *y* (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit *m* in year *y* (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year *y* (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	The most recent year for which the generation data is available

As it is difficult to obtain the detailed data on the power generation, fuel consumption and thermal efficiency of each newly built power unit from public documents, a deviation of TOOL07 (07.0) is adopted following the clarifications[12] given by the CDM EB concerning the BM emission factor calculation:

(1) The CDM EB suggested using the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin.

(2) The EB agreed the use of capacity additions during last 1 ~ 3 years for estimating the build margin emission factor for grid electricity.

[12]http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP1OZAK6V5YXPQKK7WYJ ("Request for clarification on use of approved methodology AM0005 for several projects in China", the EB's guidance on DNV deviation request)

(3) The EB also agreed to use of weights estimated using installed capacity in place of annual electricity generation.

The newly built power plants in the past few years are bundled into "grouped new power plant" according to their construction year, their province and their fuel type. The annual net electricity generation in year y of each "grouped new power plant" $EG_{m,y}$ is estimated according to their total capacity and the average utilization hours, using the following equation:

$$EG_{m,y} = CAP_m \times H_{m,y} \quad (8)$$

Where:

$EG_{m,y}$	=	Annual net electricity generation the unit *m* in year *y* (MWh)
CAP_m	=	Installed capacity of the unit *m* (MW)
$H_{m,y}$	=	Utilization hour of the unit *m* in the year *y* (h), determined according to the average utilization hour of the same type of unit in the same province
y	=	The most recent year for which the generation data is available. For the calculation of BM in 2019, *y* = 2017
m	=	grouped new power plant

Since the newly built power plants in the same province (A), in the same year (t) and using the same fuel type (k) are grouped into "a grouped new power plant", CAP_m represents the total installed capacity of fuel type *k* power plants located in the province *A* and in the year *t*:

$$CAP_m = CAP_{A,t,k} \quad (9)$$

Where:

CAP_m	=	Installed capacity of the unit *m* (MW), with *m* representing the specified combination of *A*, *t*, and *k*
$CAP_{A,t,k}$	=	Total installed capacity of fuel type *k* power plants located in the province *A* and in year *t*
A	=	Provinces covered by CCPG, namely, Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province and Chongqing City
t	=	Years related to the grouped new power plants; for the 2019 calculation, *t* represents 2017, 2016, 2015.... until the aggregated electricity generation of the grouped new power plants reaches 20% of the total electricity generation of CCPG
k	=	Fuel type of the grouped new power plants, including hydro, thermal (coal, gas, oil, waste incineration, other thermal), nuclear, wind, solar and others.

Figure 4.1 shows the procedure applied to determine the sample group of power units *m*.

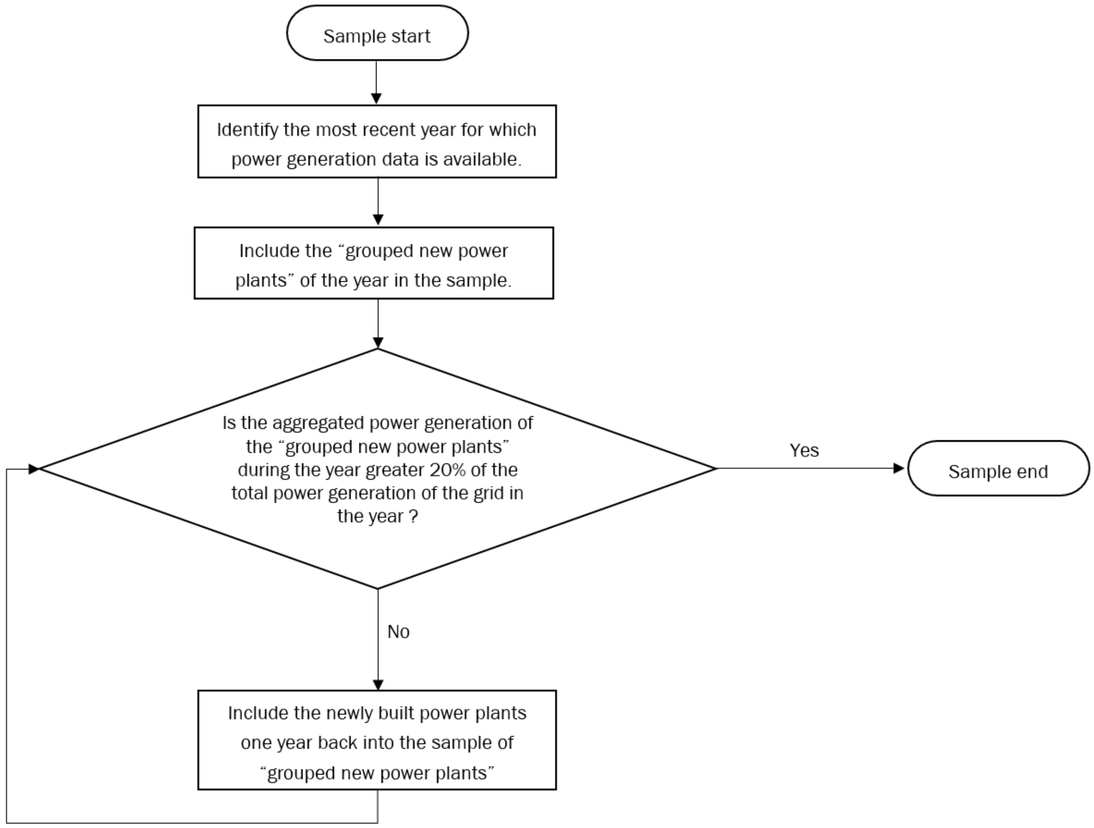


Figure 4.1 Procedure to determine the sample group of power units *m*

The emission factors of each fuel type $EF_{EL,m,y}$ are determined according to Option A2 in TOOL07 (version 07.0), as the following equation:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \tag{10}$$

Where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit *m* in year *y* (t CO_2/MWh)
$EF_{CO2,m,i,y}$	=	Average CO_2 emission factor of fuel type *i* used in power unit *m* in year *y* (t CO_2/GJ)
$\eta_{m,y}$	=	Average net energy conversion efficiency of power unit *m* in year *y* (ratio)
m	=	All power units serving the grid in year y except low-cost / must-run power units
3.6	=	Conversion factor (GJ/MWh)

Among the fuel types, the emission factors of hydro, nuclear, wind, solar, other thermal and others are 0. Concerning the emission factors of coal, gas, oil and waste incineration, Equation (10) takes the following form, to be conservative:

$$EF_{best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{best,y}} \tag{11}$$

Where:

$EF_{best,m,y}$	=	Emission factor of power unit m with the best technology commercially available in year *y* (t CO_2/MWh)
$\eta_{best,y}$	=	Power generation efficiency of the best technology commercially available in year *y*
m	=	Power units serving the grid with coal, gas, oil or waste incineration in year *y*

The emission factors of coal, natural gas, fuel oil and municipal solid waste (MSW) are sourced from *2006 IPCC Guidelines for National Greenhouse Gas Inventories*, and the lower values of the 95% confidence interval are applied to be conservative.

According to *Compilation of Statistics of the Electric Power Industry* in 2017 published by China Electricity Council, newly built coal-fired power projects of 600 MW and above totalled 19.58 GW in 2017, of which 7 units each with a capacity of 1,000 MW accounted for 35.75%, and 19 units each with a capacity between 600 MW and 1,000 MW accounted for 64.25%. Among these 26 new units, the power generation and coal consumption data for 16 units are available. The 10 units with the lowest coal consumption for power supply are selected as the best commercial technology cases, and their weighted-average coal consumption is 297.59 gce/kWh, equivalent to a power generation efficiency of 41.33%.

According to *Compilation of Statistics of the Electric Power Industry* in 2017 published by China Electricity Council, 59 newly built MSW power units totalled to 1,147.5 MW in 2017, of which the power generation and MSW consumption data for 25 units are available. Their weighted-average power generation efficiency is 516.41 gce/kWh, equivalent to 23.82%.

According to *Compilation of Statistics of the Electric Power Industry* in 2017 published by China Electricity Council, 32 newly built natural gas power units totalled to 5,658.1 MW in 2017, of which the power generation and gas consumption data for 18 units are available. Their weighted-average power generation efficiency is 223.41 gce/kWh, equivalent to 55.05%.

No fuel oil power units were constructed in 2017. The power generation efficiency of the best technology commercially available for fuel oil units comes from *Baseline Emission Factors for Regional Grids in China* of previous years, i.e., 232.3 gce/kWh or 52.9%.

The calculation result of the BM emission factor $EF_{grid,BM,y}$ for CCPG is 0.2854 tCO_2/MWh, as is confirmed by *2019 Baseline Emission Factors for Regional Power Grids in China* published by the Development and Reform Commission of China.

Step 6: Calculate the combined margin (CM) emission factor.

The combined margin emission factor $EF_{grid,CM,y}$ is calculated based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

The weighted average CM method should be used as the preferred option. The simplified CM method can only be used if the project activity is located in a Least Developed Country (LDC) or in a country with less than 10 registered CDM projects at the starting date of validation or a Small Island Developing States (SIDS), which means this method is not applicable.

Option A (weighted average CM) is applied.

The combined margin (CM) emission factor $EF_{grid,CM,y}$ of the baseline scenario is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad (12)$$

Where:

$EF_{grid,OM,y}$	=	Build margin CO_2 emission factor in year *y* (tCO_2/MWh)
$EF_{grid,BM,y}$	=	Operating margin CO_2 emission factor in year *y* (tCO_2/MWh)
w_{OM}	=	Weighting of build margin emissions factor (%)
w_{BM}	=	Weighting of operating margin emissions factor (%)

As per TOOL07, w_{OM} and w_{BM} are both 50% during the crediting period. Therefore, the baseline CM emission factor:

$EF_{grid,CM,y}$= 0.8587 tCO_2/MWh × 50% + 0.2854 tCO_2/MWh× 50% = 0.5720 tCO_2/MWh.

For the quantification of baseline emissions, please refer to Section 4.4 (Net GHG Emission Reductions and Removals).

4.2 Project Emissions

As per AMS-I.D. (Version 18.0), project emission of the project is zero.

As per AMS-III.G. (Version 10.0), project emission is calculated as per equation below:

$$PE_y = PE_{Power,y} + PE_{flare,y} + PE_{process,y} \quad (13)$$

Where:

PE_y	=	Project emissions in year *y* (tCO_2e)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (tCO_2e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the landfill gas stream in the year y (tCO_2e)
$PE_{process,y}$	=	Emissions from the landfill gas upgrading process in the year y (tCO_2e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

The project does not involve the emissions from the consumption of fossil fuels, only emissions from the use of electricity for the operation of the installed facilities are involved. The project will consume electricity delivered from the grid when the project is not in operation. And this part of electricity has already been deducted from the quantity of electricity supplied by the project plant/unit to the grid when calculating the Quantity of net electricity generation supplied by the project plant/unit to the grid, which is then used to calculate the Baseline emissions associated with electricity generation ($BE_{EC,y}$), hence emissions from the use of electricity for the operation of the installed facilities are excluded, which means Emissions from the use of fossil fuel or electricity for the operation of the installed facilities equal to 0.

The project does not claim emission reductions from flaring, hence project emissions from flaring or combustion of the landfill gas stream are excluded.

The project does not involve a landfill gas upgrading process, hence the Emissions from the landfill gas upgrading process are also excluded.

In conclusion, the project emission of the project activity is 0.

4.3 Leakage

No leakage effects are accounted for under AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0).

4.4 Net GHG Emission Reductions and Removals

The emission reduction achieved by the project activity can be estimated ex-ante as follows.

$$ER_y = BE_y - PE_y - LE_y = BE_y \qquad (14)$$

As per AMS-III.G (Version 10.0), the actual emission reduction achieved for Type III component of the project (which means not including emission reduction from Type I component, i.e. LFG power generation) during the crediting period will be calculated ex post using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times \left(F_{CH4,PJ,y} - F_{CH4,BL,y}\right) \times GWP_{CH4} - PE_y - LE_y \qquad (15)$$

Where:

$F_{CH4,PJ,y}$	=	Methane captured and destroyed/gainfully used by the project activity in the year y (tCH_4)

As per section 4.2 and 4.3 above, project emission equals to zero, and no leakage is considered for the project. Therefore, the total emission reduction by the project (both Type I and Type III components) is calculated as follows:

$$ER_{y,calculated} = (1 - OX) \times \left(F_{CH4,PJ,y} - F_{CH4,BL,y}\right) \times GWP_{CH4} + BE_{EC,y} \qquad (16)$$

In China there is no mandatory regulatory requirement on landfill gas destruction/combustion that has been systematically enforced. Neither is there any contractual requirement on methane destruction that the project owner is bound to. Therefore, $F_{CH4,BL,y}$is 0.

For project activities that utilize the recovered methane for power generation, $F_{CH4,PJ,y}$ may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$F_{CH4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4} \quad (17)$$

Where:

EG_y	=	Electricity generation in year y (MWh)
3600	=	Conversion factor (1 MWh = 3600 MJ)
NCV_{CH4}	=	NCV of methane (MJ/Nm³), use default value: 35.9 MJ/Nm³
D_{CH4}	=	Density of methane at the temperature and pressure of the landfill gas in the year y (t/m³).
EE_y	=	Energy Conversion Efficiency of the project equipment determined from one of the following options:

- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation.
- Default efficiency of 40 percent

Ex-ante calculation of emission reductions:

The estimated emission reductions (ER_y) of the project is calculated as follows:

1. Calculation of the baseline emissions

The baseline emissions are calculated as follows:

Baseline emissions resulted from Solid Waste Disposal Site (AMS-III.G).

As described in 4.1, Baseline emissions of methane from the SWDS ($BE_{CH4,SWDS,y}$) is calculated ex-ante using equation (2) and equation (3).

Table 4.1 Necessary values to calculate $BE_{CH4,SWDS,y}$

Parameter	Value	Data sources
η_{PJ}	80%	Feasibility study report （FSR）
φ_y	0.75	Emissions from solid waste disposal sites (version 08.0)
f_y	0	FSR

GWP_{CH4}	27.9		IPCC AR6
OX	0.1		IPCC 2006 Guidelines for National Greenhouse Gas Inventories
F	0.5		IPCC 2006 Guidelines for National Greenhouse Gas Inventories
$DOC_{f,y}$	0.5		IPCC 2006 Guidelines for National Greenhouse Gas Inventories
MCF_y	1.0		IPCC 2006 Guidelines for National Greenhouse Gas Inventories
$P_{n,j,x}$ (Weight fraction of waste type j in the landfill)	Pulp, paper and cardboard	4.30%	FSR
	Textiles	7.30%	
	Food, food waste, beverages and tobacco	53.43%	
	Wood and wood products	4.90%	
	Garden, yard and park waste	7.10%	
	Rubber, plastic, Glass, metal and other inert	22.97%	
	Total	100.00%	
DOC_j	Pulp, paper and cardboard	40%	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)
	Textiles	24%	
	Food, food waste, beverages and tobacco	15%	
	Wood and wood products	43%	
	Garden, yard and park waste	20%	
	Others	-	
K_j	Pulp, paper and cardboard	0.06	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)
	Textiles	0.06	
	Food, food waste, beverages and tobacco	0.185	

	Wood and wood products	0.03	
	Garden, yard and park waste	0.10	
	Others	-	
y	1-10		AMS-III.G. (version 10.0).

According to the methodological tool “Emissions from solid waste disposal sites” (version 08.0), the methane amount in landfill gas in the crediting period calculated by this project is higher than the methane amount destroyed by the project with the capacity of 2.134 MW. To be conservative, in the PD, equation (3) in section 4.1 of the PD is applied to estimate the methane amount. Considering the use of "solid waste treatment station emissions calculation tool" to calculate methane consumption and the actual amount of destruction may exist deviation, the PD refers to the parameters in the FSR: the installed capacity of 2.134 MW, annual operating hours (7,200 hours), the power generation efficiency (40.2%), methane calorific value (0.0504 TJ/tCH_4). The details theoretical installed capacity and methane destroyed as follows, details please refer to ER calculation sheet.

Table 4.2 The estimation of annual methane destroyed

Year	Methane amount estimated according to equation (3) (m^3)	η_{PJ}	Theoretical installed capacity (MW)	Actual installed capacity (MW)	Annual electricity generation (MWh)	Annual electricity supplied to the grid (MWh)[13]	Annual methane destroyed based on annual electricity generation(tCH_4)
2021	2,895,864	80%	1.297	2.134	9,335	8,775	1,667
2022	2,755,825	80%	1.234	2.134	8,884	8,351	1,587
2023	2,376,465	80%	1.064	2.134	7,661	7,201	1,368
2024	2,056,067	80%	0.921	2.134	6,628	6,230	1,184
2025	1,785,017	80%	0.799	2.134	5,754	5,409	1,028
2026	1,555,299	80%	0.696	2.134	5,014	4,713	896
2027	1,360,230	80%	0.609	2.134	4,385	4,122	783
2028	1,194,230	80%	0.535	2.134	3,850	3,619	688
2029	1,052,645	80%	0.471	2.134	3,393	3,190	606
2030	931,586	80%	0.417	2.134	3,003	2,823	536
2031	827,804	80%	0.371	2.134	2,669	2,508	477

Therefore, in this PD, baseline emissions resulting from AMS-III.G. (Version 10.0) are ex-ante estimated as follows.

[13] Annual electricity supplied to the grid = [1- Plant self-consumption rate (%)]*Annual electricity generation

Table 4.3 The ex-ante estimation of baseline emissions resulted from AMS-III.G.

Period	OX	$F_{CH4,PJ,y}$ (tCH4)	$F_{CH4,BL,y}$ (tCH4)	GWP_{CH4} (tCO2e/tCH4)	$BE_{CH4,y}$ (tCO2e)
26-Nov-2021 to 31-Dec-2021	0.1	164	0	27.9	4,564
01-Jan-2022 to 31-Dec-2022	0.1	1,587	0	27.9	44,041
01-Jan-2023 to 31-Dec-2023	0.1	1,368	0	27.9	37,978
01-Jan-2024 to 31-Dec-2024	0.1	1,184	0	27.9	32,858
01-Jan-2025 to 31-Dec-2025	0.1	1,028	0	27.9	28,526
01-Jan-2026 to 31-Dec-2026	0.1	896	0	27.9	24,855
01-Jan-2027 to 31-Dec-2027	0.1	783	0	27.9	21,737
01-Jan-2028 to 31-Dec-2028	0.1	688	0	27.9	19,085
01-Jan-2029 to 31-Dec-2029	0.1	606	0	27.9	16,822
01-Jan-2030 to 31-Dec-2030	0.1	536	0	27.9	14,887
01-Jan-2031 to 25-Nov -2031	0.1	430	0	27.9	11,924
Total		9,269			257,277

Baseline emissions resulted from power generation (AMS-I.D).

According to the project FSR, the annual operational hour of the power generation units are 7,200 hours and the captive electricity consumption by the project activity is 6% of the total power generation. The CO_2 emissions from electricity generation by grid-connected thermal power plants in the baseline scenario which are displaced by LFG electricity generation under the project activity are tabulated below. The baseline emission factor ($EF_{CM,grid,y}$) is 0.5720 tCO_2e/MWh

Table 4.4 CO_2 emissions from electricity generation in baseline

Period	$EG_{PJ,facility,y}$ Net electricity generation (MWh)	$EF_{grid,CM,y}$ (tCO_2/MWh)	$BE_{EC,y}$ CO_2 emissions from electricity generation in baseline scenario (t CO_2e)
26-Nov-2021 to 31-Dec-2021	866	0.5720	495
01-Jan-2022 to 31-Dec-2022	8,351	0.5720	4,776
01-Jan-2023 to 31-Dec-2023	7,201	0.5720	4,119

01-Jan-2024 to 31-Dec-2024	6,230	0.5720	3,563
01-Jan-2025 to 31-Dec-2025	5,409	0.5720	3,094
01-Jan-2026 to 31-Dec-2026	4,713	0.5720	2,695
01-Jan-2027 to 31-Dec-2027	4,122	0.5720	2,357
01-Jan-2028 to 31-Dec-2028	3,619	0.5720	2,069
01-Jan-2029 to 31-Dec-2029	3,190	0.5720	1,824
01-Jan-2030 to 31-Dec-2030	2,823	0.5720	1,614
01-Jan-2031 to 25-Nov -2031	2,261	0.5720	1,293
Total	48,785		27,899

Thus, the baseline emissions are determined as below.

Table 4.5 The estimation of emission reductions BE_y

Period	$BE_{CH4,y}$ Baseline emissions resulted from AMS-III.G. (tCO_2e)	$BE_{EC,y}$ CO_2 emissions from electricity generation in baseline scenario (tCO_2e)	BE_y (tCO_2e)
26-Nov-2021 to 31-Dec-2021	4,564	495	5,059
01-Jan-2022 to 31-Dec-2022	44,041	4,776	48,817
01-Jan-2023 to 31-Dec-2023	37,978	4,119	42,097
01-Jan-2024 to 31-Dec-2024	32,858	3,563	36,421
01-Jan-2025 to 31-Dec-2025	28,526	3,094	31,620
01-Jan-2026 to 31-Dec-2026	24,855	2,695	27,550
01-Jan-2027 to 31-Dec-2027	21,737	2,357	24,094
01-Jan-2028 to 31-Dec-2028	19,085	2,069	21,154
01-Jan-2029 to 31-Dec-2029	16,822	1,824	18,646
01-Jan-2030 to 31-Dec-2030	14,887	1,614	16,501
01-Jan-2031 to 25-Nov -2031	11,924	1,293	13,217
Total	257,277	27,899	285,176

2. Calculation of the project emissions

As described in section 4.2, project emission is zero.

3. Calculation of the leakage

As described in section 4.3, the leakage is not considered.

4. Calculation of emission reductions

The estimated emission reductions of the 10-year crediting period are calculated as follows:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
26-Nov-2021 to 31-Dec-2021	5,059	0	0	5,059
01-Jan-2022 to 31-Dec-2022	48,817	0	0	48,817
01-Jan-2023 to 31-Dec-2023	42,097	0	0	42,097
01-Jan-2024 to 31-Dec-2024	36,421	0	0	36,421
01-Jan-2025 to 31-Dec-2025	31,620	0	0	31,620
01-Jan-2026 to 31-Dec-2026	27,550	0	0	27,550
01-Jan-2027 to 31-Dec-2027	24,094	0	0	24,094
01-Jan-2028 to 31-Dec-2028	21,154	0	0	21,154
01-Jan-2029 to 31-Dec-2029	18,646	0	0	18,646
01-Jan-2030 to 31-Dec-2030	16,501	0	0	16,501
01-Jan-2031 to 25-Nov -2031	13,217	0	0	13,217
Total	285,176	0	0	285,176

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	GWP_{CH_4}
Data unit	tCO_2e/tCH_4
Description	Global warming potential of CH_4
Source of data	*IPCC Sixth Assessment Report*
Value applied	27.9
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	η_{PJ}
Data unit	-
Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	FSR
Value applied	80%
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	D_{CH4}
Data unit	tCH_4/m^3 CH_4
Description	Density of methane at the temperature and pressure of the landfill gas in the year y

Source of data	Default value as per the latest version of “Project emissions from flaring”
Value applied	0.716 at normal condition
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	φ_y
Data unit	-
Description	Model correction factor to account for model uncertainties for year y
Source of data	Default value from TOOL04: “Emissions from solid waste disposal sites” (version 08.0)
Value applied	0.75
Justification of choice of data or description of measurement methods and procedures applied	Default values for the model correction factor <table><tr><td></td><td>Humid/wet conditions</td><td>Dry conditions</td></tr><tr><td>Application A</td><td>0.75</td><td>0.75</td></tr><tr><td>Application B</td><td>0.85</td><td>0.80</td></tr></table>Application A (the project activity mitigates methane emissions from a specific existing SWDS) is applied, and the default value of 0.75 is thus used.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	f_y
Data unit	-

Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year *y*
Source of data	TOOL04: “Emissions from solid waste disposal sites” (version 08.0)
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is determined once for the crediting period ($f_y = f$) according to TOOL04 (version 08.0).

Data / Parameter	NCV_{CH4}
Data unit	MJ/Nm^3
Description	Net calorific value of methane at reference conditions
Source of data	AMS-III.G. “Landfill methane recovery” (version 10.0).
Value applied	35.9
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	OX
Data unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	Based on an extensive review of published literature on this subject, including the *IPCC 2006 Guidelines for National Greenhouse Gas Inventories*

Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	When methane passes through the top-layer, part of it is oxidized by methanotrophic bacteria to produce CO_2. The oxidation factor represents the proportion of methane that is oxidized to CO_2 This should be distinguished from the methane correction factor (*MCF*) which is to account for the situation that ambient air might intrude into the SWDS and prevent methane from being formed in the upper layer of SWDS.

Data / Parameter	F
Data unit	(Volume fraction)
Description	Fraction of the methane in the SWDS gas
Source of data	*IPCC 2006 Guidelines for National Greenhouse Gas Inventories*
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	Upon biodegradation, organic material is converted to a mixture of methane and carbon dioxide.

Data / Parameter	$DOC_{f,y}$
Data unit	(Weight fraction)
Description	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year *y*
Source of data	*IPCC 2006 Guidelines for National Greenhouse Gas Inventories*
Value applied	0.5

Justification of choice of data or description of measurement methods and procedures applied	$DOC_{f,y} = DOC_{f,default}$, because Application A in TOOL04 (version 08.0) is applied.
Purpose of Data	Calculation of baseline emissions
Comments	This factor reflects the fact that some degradable organic carbon does not degrade, or degrades very slowly, in the SWDS. This default value can only be used for: (a) Application A; or (b) Application B if the tool is applied to MSW.

Data / Parameter	MCF_y
Data unit	-
Description	Methane correction factor for year *y*
Source of data	*IPCC 2006 Guidelines for National Greenhouse Gas Inventories*
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	The SWDS does not have a water table above the bottom of the SWDS and Application A is applied. Therefore, a default value ($MCF_y = MCF_{default}$) is selected from TOOL04 (version 08.0) according to the management of the SWDS. Yueyang County MSW Landfill has controlled placement of waste, mechanical compacting and levelling of the waste. The default value of 1.0 for anaerobic managed solid waste disposal sites is applied.
Purpose of Data	Calculation of baseline emissions
Comments	MCF accounts for the fact that unmanaged SWDS produce less methane from a given amount of waste than managed SWDS, because a larger fraction of waste decomposes aerobically in the top layers of unmanaged SWDS.

Data / Parameter	$W_{j,x}$
Data unit	t
Description	Amount of organic waste type *j* disposed in the SWDS in the year *x*
Source of data	FSR of the Project and Annual MSW disposed record of landfill

<table>
<tr><td>Value applied</td><td>The amounts of each waste type are calculated based on actual disposal rates and estimated composition of the MSW disposed at Yueyang County MSW Landfill.
<table>
<tr><td>Year</td><td>Annual disposed MSW (tons)</td></tr>
<tr><td>2013</td><td>8,769</td></tr>
<tr><td>2014</td><td>35,223</td></tr>
<tr><td>2015</td><td>32,497</td></tr>
<tr><td>2016</td><td>36,096</td></tr>
<tr><td>2017</td><td>205,424</td></tr>
<tr><td>2018</td><td>139,462</td></tr>
<tr><td>2019</td><td>107,983</td></tr>
<tr><td>2020</td><td>107,736</td></tr>
<tr><td>2021</td><td>110,030</td></tr>
<tr><td>2022</td><td>47,931</td></tr>
</table>
<table>
<tr><td>Waste type j</td><td>Weight fraction</td></tr>
<tr><td>Wood and wood products</td><td>4.90%</td></tr>
<tr><td>Pulp, paper and cardboard</td><td>4.30%</td></tr>
<tr><td>Food, food waste, beverages and tobacco</td><td>53.43%</td></tr>
<tr><td>Textiles</td><td>7.30%</td></tr>
<tr><td>Garden, yard and park waste</td><td>7.10%</td></tr>
<tr><td>Glass, plastic, metal, other inert waste</td><td>22.97%</td></tr>
</table>
</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>-</td></tr>
<tr><td>Purpose of Data</td><td>Calculation of baseline emissions</td></tr>
<tr><td>Comments</td><td>-</td></tr>
</table>

<table>
<tr><td>Data / Parameter</td><td>DOC_j</td></tr>
<tr><td>Data unit</td><td>(Weight fraction)</td></tr>
<tr><td>Description</td><td>Fraction of degradable organic carbon in the waste type j</td></tr>
<tr><td>Source of data</td><td>TOOL04: “Emissions from solid waste disposal sites” (version 08.0)</td></tr>
<tr><td>Value applied</td><td>Default values for DOC_j
<table>
<tr><td>Waste type j</td><td>DOC_j
(% wet waste)</td></tr>
<tr><td>Wood and wood products</td><td>43</td></tr>
<tr><td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr>
<tr><td>Food, food waste, beverages and tobacco (other than sludge)</td><td>15</td></tr>
<tr><td>Textiles</td><td>24</td></tr>
<tr><td>Garden, yard and park waste</td><td>20</td></tr>
<tr><td>Glass, plastic, metal, other inert waste</td><td>0</td></tr>
</table></td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>-</td></tr>
<tr><td>Purpose of Data</td><td>Calculation of baseline emissions</td></tr>
<tr><td>Comments</td><td>The percentages listed in table above are based on wet waste basis which are concentrations in the waste as it is delivered to the SWDS.</td></tr>
</table>

Data / Parameter	k_j
Data unit	1/yr
Description	Decay rate for the waste type *j*
Source of data	TOOL04: “Emissions from solid waste disposal sites” (version 08.0), *IPCC 2006 Guidelines for National Greenhouse Gas Inventories* (adapted from Volume 5, Table 3.3)

<table>
<tr><td>Value applied</td><td>Provide the value applied</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>
The project site is located in Yueyang City, where the meteorological data[14] are:
Mean annual temperature (MAT): 17.0°C⩽ 20°C;
Mean annual precipitation (MAP): 1320.3 mm;
Potential evapotranspiration (PET): 656.9 mm, MAP/PET > 1
Therefore, the values for wet, boreal and temperate area shall be applied.
<table>
<tr><th colspan="2">Waste type j</th><th>Boreal and Temperate
(MAT⩽20°C),
and wet
(MAP/PET>1)</th></tr>
<tr><td rowspan="2">Slowly degrading</td><td>Pulp, paper, cardboard (other than sludge), textiles</td><td>0.06</td></tr>
<tr><td>Wood, wood products and straw</td><td>0.03</td></tr>
<tr><td>Moderately degrading</td><td>Other (non-food) organic putrescible garden and park waste</td><td>0.10</td></tr>
<tr><td>Rapidly degrading</td><td>Food, food waste, sewage sludge, beverages and tobacco</td><td>0.185</td></tr>
</table>
</td></tr>
<tr><td>Purpose of Data</td><td>Calculation of baseline emissions</td></tr>
<tr><td>Comments</td><td>-</td></tr>
</table>

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2/MWh
Description	Operating margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by DNA of China

[14] http://www.yyx.gov.cn/37584/38154/38157/38160/38207/38211/40174/content_1314723.html

Value applied	0.8587
Justification of choice of data or description of measurement methods and procedures applied	Official and authoritative statistic data.
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	According to section 4.1, the ex ante option is selected to calculate $EF_{grid,OM,y}$ and fixed in the crediting period.

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2/MWh
Description	Build margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by DNA of China
Value applied	0.2854
Justification of choice of data or description of measurement methods and procedures applied	Official and authoritative statistic data.
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	According to section 4.1, the ex ante option is selected to calculate $EF_{grid,BM,y}$ and fixed in the crediting period.

Data / Parameter	$EF_{grid,CM,y}$ ($EF_{grid,y}$)
Data unit	tCO_2/MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Calculated based on "Tool to calculate the emission factor for an electricity system"
Value applied	0.5720
Justification of choice of data or description of measurement methods and procedures applied	$EF_{grid,CM,y}$ is calculated based on $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ as per the latest version of "Tool to calculate the emission factor for an electricity system" (version 07.0).

Purpose of Data	Calculation of baseline emissions and project emissions
Comments	According to section 4.1, the ex ante option is selected to calculate $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ and fixed in the crediting period, thus the $EF_{grid,CM,y}$ is fixed during the crediting period.

Data / Parameter	EE_y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	Specification provided by the equipment manufacturer
Value applied	40.2%
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,facility,y}$
Data unit	MWh
Description	Net amount of electricity generated using LFG by the project activity in year *y*
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	Measured by a bi-directional electricity meter
Frequency of monitoring/recording	Continuously measured and regularly recorded
Value applied	Please refer to Section 4.4 for details
Monitoring equipment	Electricity meter with an accuracy of no lower than 0.5 S
QA/QC procedures to be applied	The monitored values will be cross checked with sales receipts. The electricity meter is to be calibrated regularly in compliance with the latest version of "*Technical administrative code of electric energy metering*".

	The calibration reports, recorded data files and sales receipts will be archived for two years following the end of the crediting period.
Purpose of data	Calculation of baseline emissions
Calculation method	This parameter should be calculated as difference between (a) the quantity of electricity supplied by the project to the grid ($EG_{export,y}$); and (b) the quantity of electricity the project from the grid ($EG_{import,y}$). The formula is as follows: $EG_{PJ,facility,y} = EG_{export,y} - EG_{import,y}$ The following parameters shall be measured: (a) The quantity of electricity supplied by the project to the grid; and (b) The quantity of electricity delivered to the project from the grid
Comments	-

Data / Parameter	EG_y
Data unit	MWh
Description	Electricity generation by the project activity in year *y*
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	Measured by a bi-directional electricity meter
Frequency of monitoring/recording	Continuously measured and regularly recorded
Value applied	Ex post calculation only
Monitoring equipment	Electricity meter with an accuracy of no lower than 0.5 S
QA/QC procedures to be applied	The electricity meter will be periodically calibrated
Purpose of data	Calculation of emission reduction
Calculation method	-
Comments	This parameter is required for calculating baseline emissions

5.3 Monitoring Plan

The monitoring plan presented in this PD assures that real, measurable, long-term GHG emission reductions can be monitored, recorded and reported. It is a crucial procedure to identify the final VCUs of the project. This monitoring plan will be implemented by the project owner during the project operation. The details of the monitoring plan are specified as follows:

（A） Data and parameters to be monitored

Data and parameters to be monitored are listed below shows the positions of the monitoring instruments and Table 5.1 list the corresponding parameters monitored:

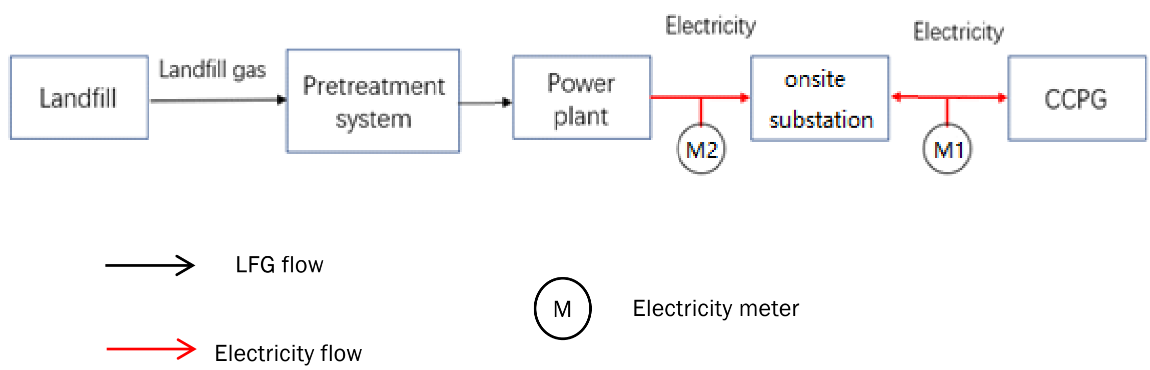


Figure 5.1 Diagram of the project boundary showing the monitored parameters

Table 5.1 Corresponding parameters monitored

Position	Parameter Monitored	Description
M1	$EG_{PJ,facility,y}$ $(EG_{PJ,y})$	Quantity of net electricity generation supplied by the project to the grid is continuously monitored by bi-directional electricity meter installed by outlet of onsite substation with the accuracy of no less than 0.5s.
M2	EG_y	Amount of electricity generated using LFG by the project activity is continuously monitored by electricity meter installed at the outlet of gas generator with the accuracy of no less than 0.5s.

(B) Management structure

Figure 5.2 shows the operation and management structure of the project monitoring team. The project proponent has appointed a specific VCS monitoring team to be responsible for data collection, supervision and calculation during the project implementation. A VCS manager takes full responsibility for the overall implementation of the project monitoring plan. The monitoring and measurement of LFG, electricity generation and consumption etc., are carried out by a few designated monitoring team members. In addition, the project proponent also appointed internal verifiers who are responsible for internal check of the measurement, collection of relevant receipts and invoices, and the calculation of the emission reductions. A monitoring and management manual of the project, which identifies detailed duties and responsibilities of each team member, has been developed and serves as the guidance for the project monitoring.

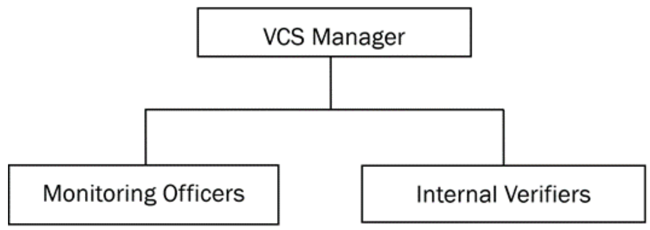


Figure 5.2 Operation and management structure of the project monitoring team

(C) Data collection

The computer system automatically monitors and records the measured data of the LFG; these automatic records will serve as the main data source for the calculation of emission reductions.

The monitoring team members are responsible for data collection and documentation. They read and archive the monitored data regularly, and ensure that all date and files are archived in time and in a proper manner.

(D) Quality assurance

All the monitoring instruments, which include the electricity meter, the flow meter and the methane content analyser, are chosen in compliance with VCS requirements, and will be calibrated regularly for accuracy by qualified third-party organizations according to the national regulations during the entire crediting period.

To assist in future verifications, the project proponent preserves the calibration records, along with the files of monitored data.

Error-checking is carried out regularly on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. In case of malfunction of any measuring instrument, the instrument supplier will provide technical support to solve the problem promptly, and the emission reductions during the troubleshooting period will be calculated conservatively.

(E) Data file management

All the monitored data are electronically files at the end of each month and the electronic data files are archived in both disk copy and printed hard copy. Other documents in paper e.g., maps, forms and EIA reports are preserved in good conditions as well. All data collected as part of the monitoring must be archived electronically and be kept at least for two years after the end of the crediting period. The project proponent will provide original records and documents if necessary.